TSM 099 Final Exam - Practice 2

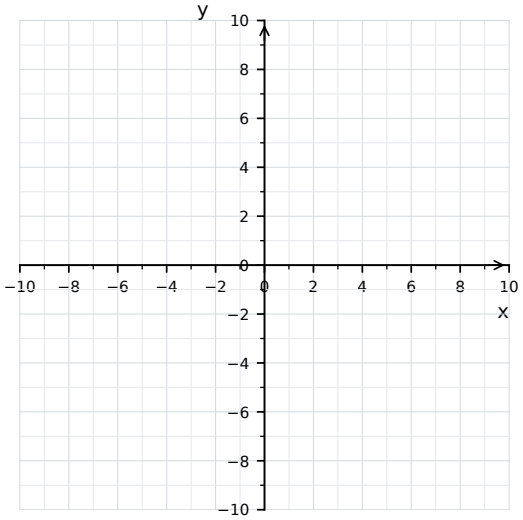
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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Find the intercepts and graph the line on the grid.

$x + 2y = 8$



2. Solve using the quadratic formula. Give exact simplified solutions.

$3x^2 = 5 - x$

3. Find the slope-intercept equation through these two points.

$(2,\ 1);\ (4,\ 3)$

4. Multiply and simplify.

$$\frac{x^2 - 9}{x^2 - 2x - 48} \times \frac{x + 6}{x^2 + 3x}$$

5. Add and simplify.

$$\frac{2}{x^2 - 3x} + \frac{3}{x^2 - 9}$$

6. Solve by taking square roots.

$3(x - 2)^2 = 48$

7. Solve by factoring.

$x^2 + 2x = 15$

8. Solve and check in the original equation.

$$\frac{10}{2x - 8} + \frac{2}{x - 4} = \frac{7}{3}$$

9. Factor completely.

$10x^2 - 11x - 6$

10. Evaluate at x = -2; show substitution.

$(x + 3)(3x - 6)$

Student answer sheet

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Question 1

Intercepts

4 pts

Graph (booklet)

6 pts

Question 2

Solutions

10 pts

Question 3

Equation

10 pts

Question 4

Expression

10 pts

Question 5

Expression

10 pts

Student answer sheet

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Question 6

Solutions

10 pts

Question 7

Solutions

10 pts

Question 8

Solution/check

10 pts

Question 9

Factors

10 pts

Question 10

Value

10 pts

Answer key

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Question 1

Intercepts	$(8,\ 0);\ (0,\ 4)$	4 pts
Graph (booklet)	$y = 4 - \frac{x}{2}$	6 pts

Question 2

Solutions	$x = -\frac{1}{6} + \frac{\sqrt{61}}{6},\ \ -\frac{\sqrt{61}}{6} - \frac{1}{6}$	10 pts

Question 3

Equation	$y = x - 1$	10 pts

Question 4

Expression	$\frac{x-3}{x(x-8)}$	10 pts

Question 5

Expression	$\frac{5x+6}{x(x-3)(x+3)}$	10 pts

Answer key

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Question 6

Solutions	$x = -2,\ 6$

10 pts

Question 7

Solutions	$x = -5,\ 3$

10 pts

Question 8

Solution/check	$x = 7$

10 pts

Question 9

Factors	$(2x - 3)(5x + 2)$

10 pts

Question 10

Value	-12

10 pts